

Abdikafar Omar

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EDUCATION

Duke University

Expected May 2027

Bachelor of Science in Computer Science, Concentration in Artificial Intelligence and Machine Learning

Durham, NC

- Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics and Proofs, Computer Architecture, Statistical Data Science, Introduction to Databases, Design and Analysis of Algorithms

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, R, SQL

Frameworks/Libraries: React, Next.js, LangChain, Streamlit, Pandas, NumPy, scikit-learn, NLTK, Matplotlib, PyTorch, Plotly

Tools/Platforms: Git, Docker, Linux, Node.js, ChromaDB, OpenAI API, REST APIs, UiPath, UiPath Orchestrator

RELEVANT EXPERIENCE

AI Automation Intern

May 2026 – Aug 2026

Great American Insurance Group

Cincinnati, OH

- Created and validated field-level ground-truth datasets to evaluate AI-powered document extraction and establish reliable reference outputs for measuring model performance.
- Designed and refined extraction prompts for ambiguous source content, identifying errors and edge cases through structured output evaluation.
- Conducted end-to-end testing of UiPath automation workflows, validating process execution and identifying failures through UiPath Orchestrator.

Software Engineering Intern

May 2024 – Aug 2024

Duke Office of Information Technology (Code+ Program)

Durham, NC

- Architected and implemented the core backend of the Scholars@Duke Opportunities Finder, an AI-powered platform matching university researchers with relevant funding and collaboration opportunities.
- Built an end-to-end semantic-retrieval pipeline with LangChain and ChromaDB, transforming faculty profiles and opportunity metadata into vector embeddings and ranking matches through contextual similarity search.
- Integrated metadata ingestion, embedding generation, vector indexing, and result retrieval into the full-stack application while converting researcher feedback into tested product functionality.

PROJECTS

Scholars@Duke Opportunities Finder

React, Python, LangChain, ChromaDB

- Continued maintaining and extending the platform for a semester after completing the internship, supporting faculty access to relevant funding and collaboration opportunities.
- Improved data ingestion, vector indexing, and semantic search with LangChain and ChromaDB to keep recommendations current and relevant.

Black Box Auditor | Live Demo | GitHub

Python, Streamlit, scikit-learn, Plotly, GitHub Actions

- Built a reproducible evaluation engine that tests multiple language models against seven prompt mutations under consistent temperature and token settings.
- Added documented safety metrics, TF-IDF and K-means behavioral clustering, CSV/JSON/PDF exports, and 14 automated tests executed through GitHub Actions.

EcoMind AI Carbon Footprint Explorer | Live Demo | GitHub

Python, Streamlit, Pandas, Plotly, GitHub Actions

- Built a scenario-analysis engine that estimates AI training compute, IT and facility energy use, and operational carbon emissions from configurable assumptions.
- Added batch analysis for up to 500 models, three-scenario sensitivity comparisons, interactive visualizations, CSV/JSON/PDF exports, and 14 automated tests executed through GitHub Actions.

LEADERSHIP & INVOLVEMENT

Co-President

Aug 2026 – Present

Duke Black Muslim Coalition (BMCo)

Durham, NC

- Lead organizational strategy, funding proposals, event execution, and partnerships with campus and regional organizations serving Black Muslim students.